EXAMPLE 9

Two-Story, 3D Unsymmetrical Building Frame - Dynamic Response Spectrum Analysis

Problem Description

This is a two-story three-dimensional unsymmetrical building frame. The structure is subjected to a seismic response spectrum along two horizontal axes that are at a 30-degree angle to the building axes. The seismic excitation is identical to the one used in Wilson and Habibullah (1992).

Geometry, Properties and Loads

The geometry of the structure is shown in Figure 9-1. The three-dimensional structure is modeled as a single frame with six column lines and five bays. Kip-foot-second units are used. Typical columns are 18"x18" and beams are 12"x24". The modulus of elasticity is taken as 432,000 ksf.

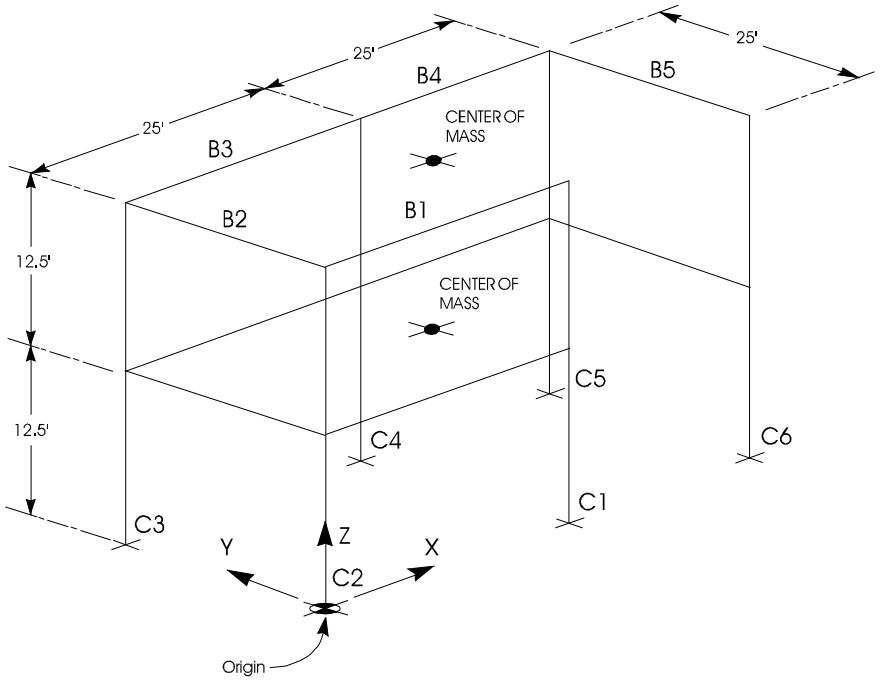

Figure 9-1 Two-Story Three-Dimensional Unsymmetrical Building Frame

Technical Features of ETABS Tested

- Three-dimensional frame analysis
- Dynamic response spectrum analysis

Results Comparison

The structure is also analyzed in Wilson and Habibullah (1992). Key ETABS results are compared in Table 9-1.

Table 9-1 Comparison of Results

Quantity	ETABS	Wilson and Habibullah
Period, Mode 1	0.4146	0.4146
Period, Mode 2	0.3753	0.3753
Period, Mode 3	0.2436	0.2436
Period, Mode 4	0.1148	0.1148
Period, Mode 5	0.1103	0.1103
Period, Mode 6	0.0729	0.0729
X- Displacement Center of Mass at 2nd Story for:		
Seismic at 30° to X	0.1062	0.1062
Seismic at 120° to X	0.0617	0.0617

Computer File

The input data file is Example 09.EDB. This file is provided as part of the ETABS installation.

Conclusions

The results comparison shows exact agreement between ETABS and the reference material.